

Journal of the American Statistical Association

ISSN: 0162-1459 (Print) 1537-274X (Online) Journal homepage: www.tandfonline.com/journals/uasa20

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To cite this article: Arnold Zellner (1992) Statistics, Science and Public Policy, Journal of the American Statistical Association, 87:417, 1-6, DOI: 10.1080/01621459.1992.10475168

To link to this article: https://doi.org/10.1080/01621459.1992.10475168

Published online: 27 Feb 2012.

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Statistics, Science and Public Policy

ARNOLD ZELLNER*

For many years, more than I would like to count, I have worked on problems involving statistics, science, and policy. Since understanding each of these subjects and their interrelations is quite difficult, some have suggested that we undertake a crash program in research to reach definitive conclusions useful, for example, to D. Allan Bromley and his client, President Bush. However, as you probably know, crash programs in research are rarely successful even though they are a lot of fun for everyone except the one paying the bills.

In connection with statistics, it's relevant to recognize the substantial growth in the number of statisticians and members and chapters of ASA during the decades of this century as shown in Figures 1 and 2. Given ASA's new constitution, the beginning of its 1991 membership drive and use of the freehand method with its optimal properties, I project that ASA membership, chapters, and sections will continue this amazing growth for many decades in the future.

Along with this growth in statistics, there has been similar growth in the "metric" fields which utilize statistics extensively, namely, econometrics, cliometrics, sociometrics, psychometrics, chemometrics, biometrics, technometrics, and so on. What has prompted this pronounced growth and what does the future hold for us? In part, this growth has occurred because, in many of the fields mentioned, there has been an effort to turn art into science using the methods of statistics, namely, experimental and survey design, estimation, testing, prediction, and other methods. To attempt to answer these questions more fully, let me review some of my experiences with statistics, science, and policy and present some of the lessons which have emerged from these experiences.

In my undergraduate and graduate study in physics years ago, we were told that in science measurements or data are fundamentally important. We did many experiments to generate data. However, we didn't receive any formal instruction in statistics to show us how to analyze and learn from our data in an effective manner. On many occasions, I used the well-known, optimal freehand method, which, as you know, produces the "right" answer all the time. For example, in analyzing my data from the Millikan oil drop experiment, it was easy to use the freehand method to fit a regression and from it the "right" electronic charge. However, many years later, my next door neighbor Jack Halpern, who has a chair in chemistry at Chicago, informed me that there were some low points in Millikan's data, about the only evidence at the time for the existence of subelectronic particles, namely, quarks. I couldn't resist saying that I had many "quarks" in my experiment. Also, I wonder what would have resulted had Millikan and I employed robust regression techniques in analyzing our data. I'm almost willing to bet that we would have gotten the "wrong" answer.

Later in research in biophysics while in the army, I had the good fortune to work in a lab which had a consulting group of statisticians, including Don Marquardt, who were kind enough to help me with analyses of my data which were subsequently published. From this and later experiences, I learned that many physical and other scientists are not well trained in statistics and on many occasions require the services of statisticians to produce and analyze their data satisfactorily. For example, in the late 1950s, I was surprised to learn that many economists were unable to perform regression analyses, let alone interpret regression results in a meaningful fashion because they were not well trained in statistics. Literally, they required help which I have provided over the years with great returns to me in the form of joint works and new ideas which stimulated my work on statistical methods. For example, it was my efforts to produce more satisfactory analyses of panel data and to simplify multivariate regression analysis that led to my multivariate "seemingly unrelated regression" estimation and testing techniques.

Many of the physical and other scientists with whom I

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In the Public Domain
Journal of the American Statistical Association
March 1992, Vol. 87, No. 417, Presidential Address

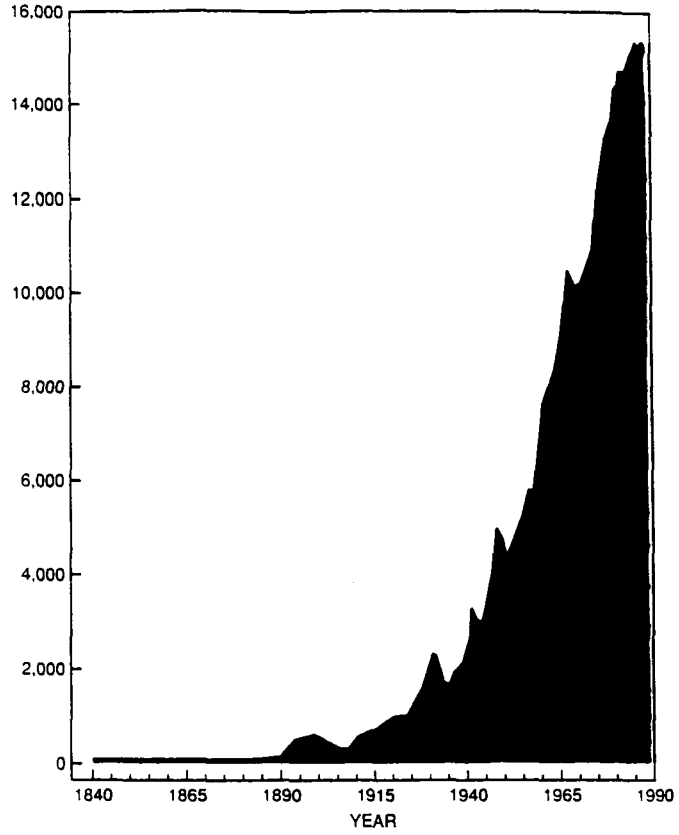


Figure 1. ASA Membership by Year. Source: Robert L. Mason, "A Golden Era of Statistics in America," in Proceedings of the American Statistical Society: Sesquicentennial Invited Paper Sessions, August 6–10, 1989 and August 22–25, 1988, 486–495.

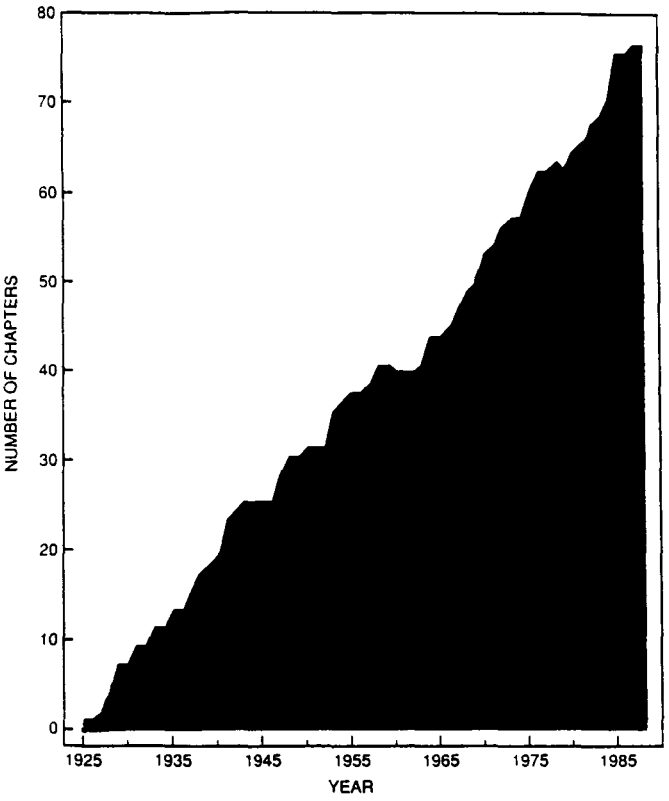


Figure 2. Number of ASA Chapters by Year. Source: Robert L. Mason, "A Golden Era of Statistics in America," in Proceedings of the American Statistical Society: Sesquicentennial Invited Paper Sessions, August 6–10, 1989 and August 22–25, 1988, 486–495.

worked over the years claimed that they deal with hard, cold, objective data. Hogwash! In a number of instances, there are problems associated with data in science as revealed by the cold fusion episode, by Millikan's notebooks, by Census undercounts, and by studies of those who report what they want to believe. As regards the cold fusion episode, the activities of Kevin Wolf, who is referred to in *Science*, October 1989, as "a hard-headed physicist [who] is skeptical about cold fusion," are described as follows:

> Wolf, among others, is looking to get good, statistically valid data on what nuclear products may be produced in the fusion cells. "That's what is lacking in this whole business—statistics," he says. Because the Pons/Fleischmann effect, if it exists, is so hard to reproduce, researchers have generally spent most of their time and money just trying to get cells that work. They generally have not set up control cells . . .

The article goes on to explain how Wolf's group and another group have done experiments ". . . with a proper blank control—cells with regular water in place of heavy water so that no . . . fusion is possible," as a check on the palladium possibly being contaminated with tritium. It's clear that in this instance producing uncontaminated data is far from a routine process.

In many other cases and fields producing good data is not usually a routine, objective, reproducible process. Thus, always, it is wise to heed the advice of Alan Enthoven, a former Undersecretary of Defense, who strongly advised us in a Chicago workshop meeting to GET THE FACTS! He made his point vividly by pointing out that in the 1950s it was estimated and believed that the Russians had twelve million military personnel in the field. During these cold war years, our Secretary of State threatened to drop the bomb on the Russians while the Europeans were singing better red than dead. At the end of the decade, someone asked how much of Russian GNP it would take to support their twelve million military personnel. After a little arithmetic and investigation, he debunked the twelve million figure by pointing out that many of the Russian divisions were reserve divisions and their division size was about seventy-five hundred rather than eleven thousand, as had been assumed. With these and other corrections, the estimated number was brought down to three to four million, however not before almost a decade of a cold war that could have turned into a hot, nuclear war at any time. Thus, then as now, the advice GET THE FACTS has great relevance for statistics, science and policy, and, fortunately, many statisticians are in the forefront in producing reliable facts. Perhaps this cold war episode points to a need for a new ASA section which would develop methods for measuring and monitoring the accuracy of news reported in the media. Certainly schools of journalism need good statistics courses.

In this connection, a 1989 cartoon in *Pravda* showed the Soviet public roaring in laughter on viewing the official Soviet economic statistics indicating strong growth and little inflation. Also, according to Chuck Waite of our Census Bureau, when Soviet government workers were told about U.S. procedures for releasing sensitive economic statistics,

one of them asked, "How can the president change a number if he doesn't like it?"

With our own official government statistics, it's hard to know what the U.S. saving rate is—see, for example, Judge (1989). Based on Department of Commerce personal savings data, the saving rate is in the vicinity of 5%. On the other hand, individual savings as reported by the Federal Reserve Board yields a much higher savings rate, in the vicinity of 8 to 9%. With other reasonable adjustments, some estimate the savings rate to be in the vicinity of 15%. This is but one example of many others pointed out by Michael Boskin and Donald Ratajczak, our excellent 1991 B&E Section Forecast Luncheon Speaker, of weak spots in our national data base that make analysis, forecasting and policy-making extremely difficult. Even though our data are much, much better than the Soviets' data, we must work hard to improve them.

As another example illustrating the importance of getting the facts, some years ago an economist did a theoretical analysis of a famous marine conservation program, the International Pacific Halibut Conservation Program set up under a treaty between Canada and the United States and studied for several years by James Crutchfield and myself (1963). In his analysis, he assumed a constant equilibrium fish population and criticized the program for not charging fishermen a fee for fishing for halibut in the North Pacific. While his theoretical argument is probably valid under some conditions, this bright analyst forgot to GET THE FACTS. The facts, carefully collected over the years by marine biologists and the statistician Douglas Chapman, clearly revealed that the halibut population was declining for many decades and not equal to a constant equilibrium value. Here indeed is an example of what Thomas Huxley has called a tragedy of science, the destruction of a beautiful hypothesis or analysis by an ugly fact.

I have become convinced over the years that we should work harder to produce more ugly facts. One way to do this is to push current theories, hypotheses, models, beliefs, et cetera in any area of science, to extremes, deduce their implications and predictions, and then do the dirty work by producing the ugly facts to show that they are outlandish. The coup de grace can be applied most effectively using statistical expertise and finesse.

In addition to thinking of statistical and other procedures to produce ugly facts rather than just dull, run-of-the-mill facts, note that the great mathematician Hadamard (1945) surveyed his fellow mathematicians and found from their responses that recognition of unusual facts played a key role in producing major breakthroughs in mathematics. While Hadamard's description of this process is fascinating, I shall just remark that recognition of a new, unusual fact prompts many minds to work to produce explanations of it, that is, new hypotheses and theories. There are many well-known examples of this process in all fields of science. Thus a statistical problem is how to design investigations to produce unusual facts rather than humdrum, old, dull facts. Some years ago, I suggested a number of rather obvious ways to produce unusual facts, namely, study unusual historical periods, survey unusual groups, look for regularities in data, collect data on incorrect predictions and forecasts of scientists and their models, study equations and properties of large-scale econometric, global warming, macromolecular and other complicated models (almost every equation of such econometric models contains at least one unusual fact), and so on—see Zellner (1984, pp. 8–10). Having produced an unusual fact, a statistician may think the job is done. If so, a glowing opportunity is being missed, namely, the creation of an explanation or hypothesis to explain the unusual fact, which, if supported by much additional data, could lead to a Nobel Prize, if only there were a Nobel Prize in statistics. In my opinion, the Nobel Prize Committee needs educating on the importance of statistics in science and policy-making and how to distinguish between loose mathematicians and fine statisticians.

Speaking of Nobel Prizes, I must mention the great empirical economic statistical work of the Nobel Prize winner Simon Kuznets. He is credited with the basic research that led to our national income and product accounts, data which are invaluable to researchers, forecasters, and private and public policy-makers. He and others recognized the importance of measurement and description in science and policy. Thus, as a graduate studuent in economics I was shocked to learn that Kuznets's, Burns's, Mitchell's, and others' work on measurement of key economic variables was damned as "measurement without theory" by a leading, theoretically oriented econometrician. Later, I coined the phrase *theory without measurement* to denigrate the irrelevant work of some theorists and also to emphasize that in all fields both theorists and empirical workers have produced important research contributions. Instead of knocking heads much of the time, it would be better for theorists and empirical workers to interact more fruitfully in statistics and other sciences.

The usefulness of such interaction is evidenced by the research of Nobel Prize winners Milton Friedman, Merton Miller, Theodore Schultz, and George Stigler, my colleagues in economics and business at the University of Chicago. All of them have done outstanding work in establishing what are the facts in their fields, analyzing the facts using modern statistical methods and relating their results to economic theory and policy. Their work was enriched by close contact with econometricians and statisticians in our Business School and University Departments of Statistics and Economics in research workshop meetings, private consultations, and Chicago luncheon discussions. With a large statistics and econometrics group in our university, fruitful interactions between members of this group and applied empirical workers and subject matter theorists take place frequently and benefit all concerned. Applied workers improve their analyses, theorists learn how to relate their theories to data, and exposure to new problems often causes statisticians and econometricians to produce new statistical and computational methods.

Further, fine work in the quality management and improvement area in our Business School involving the statisticians George Easton, Edward George, William Golomski, Rob McCulloch, Harry Roberts, George Tiao, Ruey Tsay, and others has benefited from interactions with our

School's management scientists, economists, behavioral scientists, and other specialists. In this area, the facts have been assembled and analyzed using modern time series and other techniques with important results that have, along with results of many other researchers, had a significant impact on U.S. industry. Coupled with Deming's, Taguchi's, Box's, and others' methods and philosophy, this work has resulted in a growing appreciation of statistics and statisticians. I believe that these examples of the key role of statistics and statisticians in university research are not specific to Chicago and should be recognized generally by universities, especially those without departments of statistics. In this day and age, it's my view that a university without a department of statistics is like an automobile without a motor or a soldier without a rifle.

To return to the facts, usual, unusual, ugly, et cetera that are inputs to science, what are some of the objectives of science? I have come to agree with Karl Pearson (1938) and Harold Jeffreys (1988) that a major objective of science is learning from our data and experience and another is solving practical problems. Central in the learning process and in solving practical problems are the methods of statistics, which are at the heart of the unity of science. That is, similar statistical methods are not only used in many different fields of science but also use of good statistical methods for learning from data and experience is necessary for a field to be termed scientific. This central role of statistics in science involving data production, description, design of experiments and surveys, estimation, testing, prediction, risk analysis, and decision making has come to be widely appreciated and thus, in large part, the growth of statistics, econometrics, and all the other "metrics" fields in the decades of this century.

When it is stated that learning from our data and experience is a fundamental objective of science, one may ask, "How do you learn?" Some learn informally, while Bayesians use Bayes's theorem as a formal learning model. See, for example, Jeffreys (1957, 1988), Berger (1985), Box and Tiao (1973), DeGroot (1970), Hill (1974), Press (1989), and Zellner (1971). For years, many, including members of our NSF-NBER Bayesian Seminar group, have compared results of Bayesian and non-Bayesian analyses of specific problems. I have concluded from this and other work that the Bayesian learning model is effective for use in many, frequently encountered problems and yields rather good results. This is not to say that the Bayesian model is perfect—no model in science is perfect—but just that it has been shown to be quite useful in analyzing facts with given models and in making decisions in many different fields of science. Perhaps Good's (1991) good Bayes/non-Bayes compromise put forward for many years and reviewed in an invited paper presented at this meeting will be successful in uniting Bayesians and non-Bayesians. For such an accomplishment, Good deserves the Nobel Peace Prize.

At this point, I cannot resist saying a few words about use of Balanced Loss Functions (BLFs) which will, in my opinion, create more balance between Bayesians and non-Bayesians. Consider the n-means problem, $y_i = \theta_i + \varepsilon_i$, $i = 1, 2, \ldots, n$, where θ_i is the mean of y_i, or in vector notation, $y = \theta + \varepsilon$. A BLF for this problem is $L(\theta, \tilde{\theta}) = w(y - \tilde{\theta})'(y - \tilde{\theta}) + (1 - w)(\tilde{\theta} - \theta)'(\tilde{\theta} - \theta)$, where $\tilde{\theta}$ is some estimate of θ and w is a given weight, $0 \leq w \leq 1$. The first term of this BLF reflects goodness of fit and the second, precision of estimation. Given that a posterior density for θ is available, it can be used to obtain posterior expected loss, $E_\theta L(\theta, \tilde{\theta})$, which can be minimized with respect to $\tilde{\theta}$ to obtain the minimum expected loss (MELO) estimate, $\tilde{\theta}^* = wy + (1 - w)\bar{\theta}$, where $\bar{\theta}$ is the posterior mean of θ. Currently, many non-Bayesians use $w = 1$ while most Bayesians use $w = 0$. Why not be balanced and use the MELO estimate based on a reasonable value of w which balances goodness of fit and unbiasedness and precision of estimation? For posterior expected losses, risk functions and Bayes risks of $\tilde{\theta}^*$, y and $\bar{\theta}$, see my recent paper (1991), which also considers estimation of regression coefficients.

After facts or data are assembled and described, a fundamental problem is present. What model shall we utilize in analyzing the data? In this connection, I have become enamored with the precept KISS, *k*eep *i*t *s*ophisticatedly *s*imple, not as they say in industry, *k*eep *i*t *s*imple *s*tupid because some simple models are stupid. I can think of many sophisticatedly simple models in science that work well in explanation and prediction and have challenged many audiences to tell me about a single complicated model in any field of science that works well in explanation and prediction. As yet, I have not heard of any but have discovered a number of religious believers in mind-boggling, complex models and methods. These people need saving!

Usually, given the theoretical and judgmental information that we have, an initial, tentative sophisticatedly simple model can be formulated and analyzed. When little information is available, the initial model or benchmark model might be a random walk model or some such simple model as suggested by Harold Jeffreys, Milton Friedman, and others. If there is a need to capture much detail, a model may have to be large but with care it can be sophisticatedly simple. Many large econometric and other models on the scene involve hundreds of nonlinear stochastic difference equations. It's not clear whether these models have one or many solutions and the dynamic properties of these black-box monstrosities are not well understood. Years of effort and much money have been spent to build these Edsels and more is being spent to try to salvage them. Since they are dangerous to use in investigating scientific questions and making policy, I recommend that we export them to Saddam Hussein and hope that he uses them.

The problem of how to formulate models to explain past data, to predict new data successfully, and to be useful in policy-making is a first-order problem in science. From my experience in research, consulting on several industry and governmental model-building projects, and innumerable doctoral dissertation projects, I believe that we need a good *How to Build a Model* manual. Currently much multivariate model-building is still an art, not a science. If the manual could show us how to build multivariate models as successful as the Model T Ford, it would be a best seller. However, until it appears, the advice KISS is relevant and has been successfully utilized in many fields of science.

An ingenious approach to model-building is the maximum entropy (maxent) approach used in physics and linked to statistics by Shannon, Jaynes, and others. Jaynes's (1991) invited presentation at this ASA meeting provides a number of successful applications of the approach to statistical problems. In the maxent approach a model for the observations which maximizes entropy or minimizes cross-entropy subject to side conditions reflecting given initial information is recommended since it is a conservative choice of model which incorporates the initial, available information. In addition, it has been used to produce diffuse and informative prior densities for models' parameters. Also, in working on the problem of how to produce the Pearson system of densities by maxent several years ago, it occurred to me that perhaps Bayes's Theorem could be produced as a maxent solution. With a little effort, it turned out the Bayesian learning model, that is, Bayes's Theorem, could indeed be produced as the solution to an entropy optimization problem—see Zellner (1988). Thus maxent can produce models for observations, prior densities and a learning model with its associated posterior and predictive densities. This appears to me to be a fruitful, formal framework for model-building, learning from data, and making predictions. I hasten to add that using a model to make predictions and checking their accuracy is critical in appraising the quality of models and their associated prior information, as Geisser (1980, 1991) has emphasized. This is a direct way of evaluating the quality and usefulness of different prior information embedded in alternative models and their associated prior densities as emphasized by serious model-builders and statisticians. That is, the proof of the pudding is in the eating and it is noteworthy that more and more delicious pudding has been produced in recent decades.

On policy uses of statistics, producing data to describe and monitor problem areas such as human rights violations, discrimination, the environment, business cycles, cancer incidence rates, fish populations, et cetera can help considerably to characterize problems, further understanding of them, and appraise the costs and possible outcomes of alternative policies designed to achieve what are considered to be desirable objectives. In the case of the International Pacific Halibut Conservation Program, which I mentioned earlier, this is exactly what happened with the institution of a joint Canadian–U.S. conservation program in 1923. By establishing a sensible catch limit, based on much information and a good understanding of the halibut fishery and industry, the halibut population in the North Pacific grew considerably. By the 1960s, the annual catch was about 50% larger than in the 1920s, all achieved with minimal interference in industry operations. This relatively successful conservation program (note that the Atlantic halibut population was decimated) is a good example of how initial, extensive statistical information and subsequent statistical monitoring of a program's operation, coupled with a good, relatively simple behavioral model and sensible implementation policies, can lead to reasonably successful outcomes. Perhaps some of our educational reformers who are dealing with other kinds of fish would benefit from study of the halibut conservation program.

On communication of results by statisticians to the public and policy-makers, I believe that often there are serious problems. For example, consider the following exchange between a statistician and the President of the United States:

Statistician: President, my forecast of the rate of growth of real GNP for next year is 3.0% with a 1.0 percentage point margin of error.

President: What the hell is a 1.0 percentage point margin of error?

Statistician: It means that a 90% prediction interval stretches from 2.5 to 3.5 percent growth.

President: Oh, you mean that the probability is .90 that next year's growth rate is between 2.5 and 3.5.

Statistician: Well, a Bayesian would say it that way but I mean that an interval computed over and over again from repeated samples of data would have a probability of .90 of including the random variable, next year's growth rate.

President: You must be joking. There's only one sample of data. I'll stick with the .90 figure that it's between 2.5 and 3.5 even though I'm not sure what you mean by probability. Why don't you guys talk so that I can understand you?

This example involves a conditional probability statement, the one that the President opted for and a probability statement about the operating characteristics of a procedure for computing prediction intervals. Knowing when to use one or the other of these probability statements in connection with various types of intervals, point estimates and forecasts, test statistics, et cetera has caused problems for students and practitioners for years. Further, see Jeffreys (1988, Ch. 7) for a brilliant discussion of alternative definitions of probability. Certainly, if we statisticians can't be understood, few will listen to us.

To conclude, statistical data, reasoning, and methods have come to be employed widely in all sciences and in policy-making. The number of theoretical and applied statisticians has grown markedly over the years, and many of them have assumed leading roles in academia, government, and industry. To have these favorable trends continue in the future, we must work hard to improve the quality of statistical data and methods for communicating, storing, and displaying them. There has to be a greater emphasis on the production of ugly and unusual facts and explanations of them. More work has to be done on methods for learning from data and on building models which work well in explanation and prediction and are useful in policy-making. In all these efforts and in applied work on data production, forecasting, marketing analyses, censuses and surveys, quality management and improvement, drug testing, legal statistics, and so on, if more statisticians want to assume leadership roles, I believe that there has to be more investment not only in the study of statistics but also in relevant subject matter fields, for example, medicine, biology, engineering, business, law, et cetera. Along with many others, I view the future as being bright for statisticians and particularly bright for those who have solid training in statistics and a subject matter area. Finally, the American Statistical Association has to be given a good deal of credit for its con-

structive role in fostering the successful development of statistics in the past and present. Let us all join in a toast to ASA and in an effort to make it even more successful in the future.

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